

5. Almost all cars rely on fossil fuels as their source of energy. However with supplies of fossil fuels decreasing and concerns over climate change increasing, it is important to make cars as energy efficient as possible. One type of car which is very efficient is a hybrid electric vehicle, which has both a conventional engine and an electric motor which runs from batteries. These hybrid vehicles use regenerative braking where some of the kinetic energy transferred when the brakes are applied is used to charge the batteries rather than all being lost as heat. This has a maximum energy transfer of 60%.

Data about a hybrid electric / petrol car is given below.

Time taken to accelerate from 0-100 km/h (0-28 m/s)	11.8 s
Mass of car	1 160 kg
Mean CO ₂ emissions	8.9×10^{-2} kg/km
Mean fuel economy	31.6 km/litre

(a) The car uses 10 litres of fuel per week. Calculate the mean mass of CO₂ emitted by the car in a year (52 weeks). [3]

Mean mass of CO₂ = kg

(b) Energy can be lost from cars in a variety of ways such as inertial losses, rolling resistance losses and idling losses. State the design features that manufacturers can use to reduce **each** of these energy losses. [3]

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- (c) During a journey the car slows down from 30 m/s to 10 m/s.
Use an equation from page 2 to calculate how much electrical energy is transferred to the battery if the regenerative braking process is at its maximum energy transfer. [4]

Energy transferred = J

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END OF PAPER

